GEOMETRY · RIGHT TRIANGLES AND TRIGONOMETRY

Trig Ratio Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Trig Ratio Explorer** on Mr. K's Classroom: Geometry → Right Triangles and Trigonometry → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. If you double a right triangle's size without changing its angles, does sin θ change?

   No. The ratio of the sides stays the same.

2 Explore use the tool and record what you see

Ratios at different angles

θ	sin θ	cos θ	tan θ
30°	**0.5**	**0.866**	**0.577**
45°	**0.707**	**0.707**	**1**
60°	**0.866**	**0.5**	**1.732**
35°	**0.574**	**0.819**	**0.700**

Same angle (35°), different sizes

Hypotenuse	Opposite	Adjacent	sin 35°
4	**2.29**	**3.28**	**0.574**
8	**4.59**	**6.55**	**0.574**
12	**6.88**	**9.83**	**0.574**

Special right triangles

Use the second tab.

Triangle	Short leg	Other leg	Hypotenuse
45°-45°-90°	3	**3**	**3√2**
30°-60°-90°	4	**4√3**	**8**
30°-60°-90°	5	**5√3**	**10**

3 Write the rule fill in each blank from what you observed

1. $\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$, $\cos\theta = \frac{\textbf{adjacent}}{\textbf{hypotenuse}}$ and $\tan\theta = \frac{\textbf{opposite}}{\textbf{adjacent}}$.

2. Changing the size of the triangle **does not change** the trig ratios, because the triangles are similar.

3. $\sin\theta = \cos($**90° − θ**$)$.

4. In a 45°-45°-90° triangle, hypotenuse = leg × **√2**. In a 30°-60°-90° triangle, the long leg = short leg × **√3**.

4 Check no tool: show your work

1. A right triangle has legs 6 and 8. Find the hypotenuse and the sine of the angle across from the 6.

   10; $\frac{6}{10} = \frac{3}{5}$

2. A tree casts a 20 ft shadow when the sun is 40° above the ground. How tall is the tree?

   20 tan 40° ≈ 16.8 ft

3. A 30°-60°-90° triangle has hypotenuse 14. Find both legs.

   7 and 7√3

4. Solve $\sin 28° = \frac{x}{15}$.

   x = 15 sin 28° ≈ 7.04

5 Reflect

Why does every 35° right triangle have the same sine?

All 35° right triangles are similar, so their side ratios, including $\frac{\text{opposite}}{\text{hypotenuse}}$, are equal.